

# Fluid Flow Through Fluidized Beds

Final Lab Report  
Unit Operations Lab 1  
February 2, 2026

Group 1  
*Joshua Olusoji*  
*Omar Zaruk*  
*Sanad Abu Hweij*  
*Qussay Suleiman*

**1.Executive Summary (10 pts)**

The objective of this experiment was to study how sand and ion-exchange resin behave in a gas–solid fluidized bed by measuring how the pressure drops across the bed changes with increasing and decreasing superficial air velocity. Runs were performed at two packed bed heights (6 in and 10 in) for each material, and the pressure drop per unit of bed height was recorded over a range of air velocities to identify the onset of fluidization. A sieve analysis was also completed for both materials to determine particle size distributions and calculate average particle diameters that were used in later modeling.

The sieve analysis showed that the sand was a finer material with an average particle diameter of 204.7  $\mu\text{m}$ , with most of the mass in the 200–250  $\mu\text{m}$  range. The ion-exchange resin was significantly larger, with an average particle diameter of 585.3  $\mu\text{m}$  and most of the mass near  $\sim 600 \mu\text{m}$ . Pressure-drop curves for both materials followed the expected trend:  $\Delta P/L$  increased in the fixed-bed region, then leveled off near incipient fluidization, with a small hysteresis between increasing and decreasing flow. The resin plots showed a cleaner, sharper transition and gave a minimum fluidization velocity of  $\sim 0.21 \text{ m/s}$  for both bed heights, while the sand transition was more gradual and gave  $U_{\text{mf}}$  values of  $\sim 0.07 \text{ m/s}$  (6 in) and  $\sim 0.09 \text{ m/s}$  (10 in).

Overall, the resin results matched packed-bed and fluidization behavior more closely than sand because the resin beads are more uniform and closer to spherical, which also improves agreement with the Ergun model in the fixed-bed region. The sand showed larger deviations from the Ergun prediction and a less distinct fluidization point, which is consistent with irregular particle shape, broader size distribution, and greater sensitivity to packing and flow distribution in the column.

**2. Introduction (10 pts)**

In the experiment, the behavior of a fluidized bed was studied by measuring how the pressure drop across a particle bed changed with increasing and decreasing air velocity. When air flows upward through the sand and ion-exchange resins at low velocities, the materials remain in the same position and the bed acts as a packed bed. As the air velocity increases, the resistance of air to flow through the bed increases, causing the pressure drop to get higher. At some point with enough velocity, we can observe movement of the materials and that is because we have reached this level where the flow is high enough to move the weight of the material, and this is the point where fluidization starts to occur and we can call that point the minimum fluidization velocity

The pressure drop across the bed was used to understand fluidization behavior. At low air velocities the pressure drop is not that high, but pressure drops will increase with increasing velocity, and the Ergun equation can be used to describe this behavior. As the air velocity continues to increase, a point is reached where the air is strong enough to lift the particles in the bed. At this velocity, the particles begin to move, and the bed becomes fluidized. After this point, increasing the air velocity does not cause a large increase in pressure drop

The purpose of this experiment was to study the fluidization behavior of two different solid materials which were the sand and ion exchange resin under the same conditions. Pressure drop measurements were taken at 6 inches and 10 inches, over different kinds of air velocities. A sieve analysis was performed to determine the particle size distribution and average particle size of each solid. These results were used to determine the minimum fluidization velocity and compare experimental data with theoretical values.

### 3. Theory (10 pts)

Much of the theory behind fluid flow in fluidized beds revolves around the study of transport phenomena. The transfer of momentum allows one to predict the movement of a fluid through a complex system, which is particularly important for studying fluidized beds. Because catalyst pellets come in many geometries and provide different reactions, deriving general equations to study the flow of materials in these cases is important. For this experiment, the main equation employed to study the effects of fluid flow through the fluidized bed is the Ergun Equation.

The Ergun Equation is a derivation of a momentum balance placed around the packed/fluidized bed. This equation helps relate many properties of the bed, including average pellet size, void fraction, bed height, and more to the fluid's velocity and the pressure drop across the bed. The equation can be seen below:

Where  $\Delta P$  is the pressure drop,  $L$  is the bed height,  $\rho_f$  is the fluid density,  $u$  is the superficial velocity,  $\epsilon$  is the void fraction,  $\mu$  is the fluid viscosity,  $\phi$  is the sphericity, and  $d_p$  is the particle size. The derivation begins from understanding the friction experienced by the fluid inside the fluidized bed through the “friction factor” [2]:

This is derived by executing the friction force,  $\tau_w$ , across a cylindrical tube. From here, the momentum balance can be used to find that:

Which is the representative pressure drop across a single “tube” in the Tube Bundle Model of packed beds [2]. Combining (2) and (3) and defining Reynolds Number:

Equations (2) and (4) combined describe the pressure drop in a packed bed. However, from experimental procedures, scientists have found numerical values for the laminar and highly turbulent flow of fluids through fluidized beds which when superimposed, simplify the expression into the Ergun Equation.

For this experiment, determining the superficial velocity is vital for calculations in the data analysis. The experimental superficial velocity can be calculated by taking the volumetric flowrate of air through the fluidized bed column and converting it to velocity by using the cross-sectional area of the column. To get the analytical superficial velocity, a variant of the Ergun Equation at the point of incipient fluidization can be used [3]:

This relation allows one to analytically calculate the superficial velocity and compare the values to those found experimentally.

Another important aspect for studying various pellets inside a packed bed is understanding and calculating the Average Pellet Diameter,  $\bar{d}_p$ . Because pellets of the same type tend to still have varying geometries and size, calculating the average diameter distribution of the pellets allows one to make more accurate calculations. This can be found by:

Where  $w_i$  is the mass fractions of particles of size  $d_i$  [1]. These values can be found by conducting a sieve analysis through sieves of varying pore sizes.

Lastly, it is desirable to experimentally determine the Minimum Fluidization Velocity to understand when pressure drop in the packed bed column is no longer proportional to the velocity. This can be done graphically as shown in Figures 1 and 2 below:

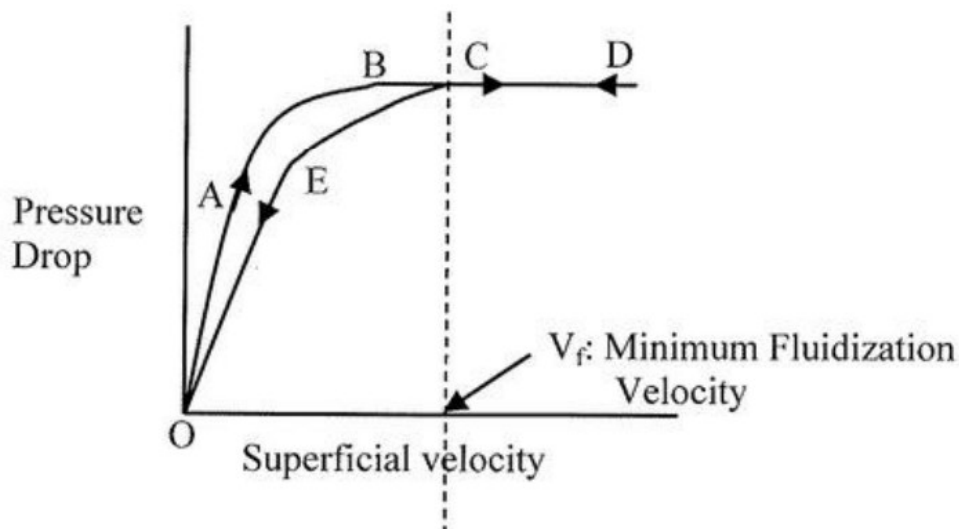


Figure 1 - Pressure Drop vs. Superficial Velocity

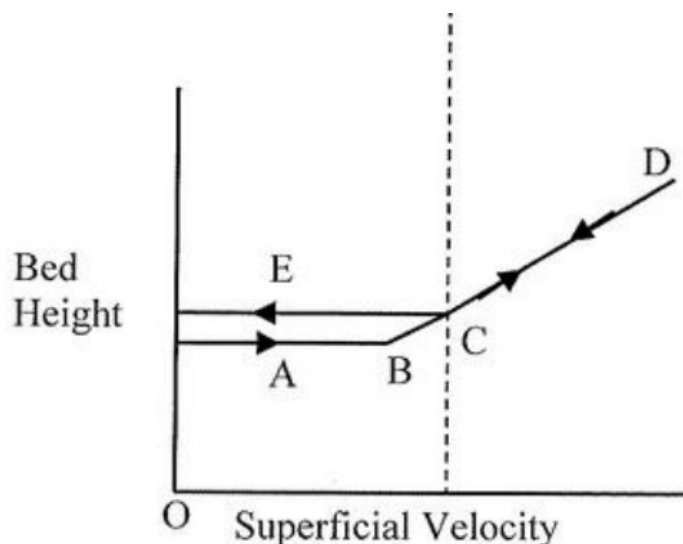


Figure 2 - Bed Height vs. Superficial Velocity

Experimental values of bed height and pressure drop can be plotted against experimental values of superficial velocity, which can be found from converting the flowrate of air in liters/minute into m/s by using the cross-sectional area of the packed column. The Minimum Fluidization Velocity is essentially when the drag forces of air become equal to those of gravity acting on the pellets [1]. At this point, no extra amount of air will change the pressure drop experienced in the system by the packed-in pellets.

The following table details the variables and their complimentary units:

Table 1 – Variables & Units

| Variable                          | Units                    |
|-----------------------------------|--------------------------|
| Pressure (P)                      | Pascals (Pa)             |
| Length (L)                        | Meters (m)               |
| Velocity (V)                      | Meters/Second (m/s)      |
| Viscosity ( $\mu$ )               | Pascals * Seconds (Pa*s) |
| Density ( $\rho$ )                | Kilograms/ (kg/)         |
| Sphericity ( $\phi$ )             | Dimensionless            |
| Particle Diameter (D)             | Meters (m)               |
| Void Fraction ( $\epsilon$ )      | Dimensionless            |
| Friction Factor (f)               | Dimensionless            |
| Radius (R)                        | Meters (m)               |
| Pressure Drop (dP or $\Delta P$ ) | Pascals (Pa)             |

#### 4. Results (15 pts)

The goal of this experiment was to study how sand and ion-exchange resin behave in a fluidized bed by examining how the pressure drop across the bed changes with increasing air velocity. Before analyzing the fluidization behavior, a sieve analysis was performed to better understand the particle size characteristics of each material. The particle size information obtained from this step was then used in later calculations, including the Ergun equation and the determination of minimum fluidization velocity.

##### Sieve Analysis and Particle Size Distribution

Sieve analysis was carried out for both the sand and ion-exchange resin samples to determine how the mass of each material was distributed across different particle sizes. The mass retained on each sieve was converted to a mass fraction and plotted against particle size to visualize the particle size distribution.

Figure 3 shows the mass fraction versus particle size for the sand sample. Most of the sand mass is concentrated in the smaller particle size range, with the highest mass fraction occurring around the 200 to 250  $\mu\text{m}$  range. Only a small amount of sand appears at the larger particle sizes, suggesting that the sample is made up primarily of fine particles with a relatively narrow distribution.

Figure 4 shows the mass fraction versus particle size for the ion-exchange resin. In contrast to sand, the resin particles are significantly larger, with most of the mass concentrated near approximately 600  $\mu\text{m}$ . The distribution is broader and shifted toward larger particle sizes, which is consistent with the uniform, manufactured shape of the resin beads.

The average particle diameter for both materials was calculated using the sieve analysis data and used in subsequent pressure drop and minimum fluidization velocity calculations. As expected, the resin particles have a larger average particle size than the sand, which contributes to the differences observed later in their fluidization behavior.

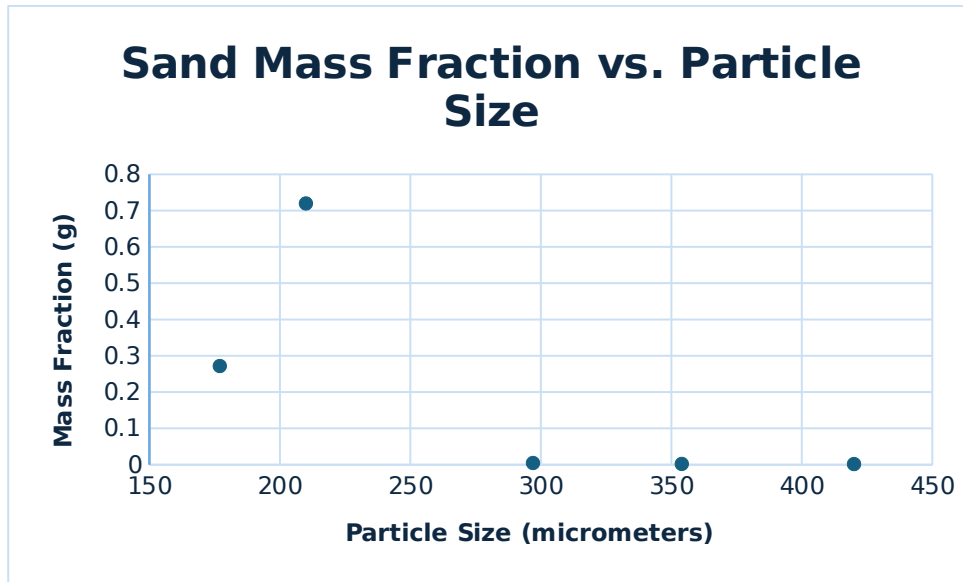


Figure 3- Sand Mass Fraction vs. Particle Size

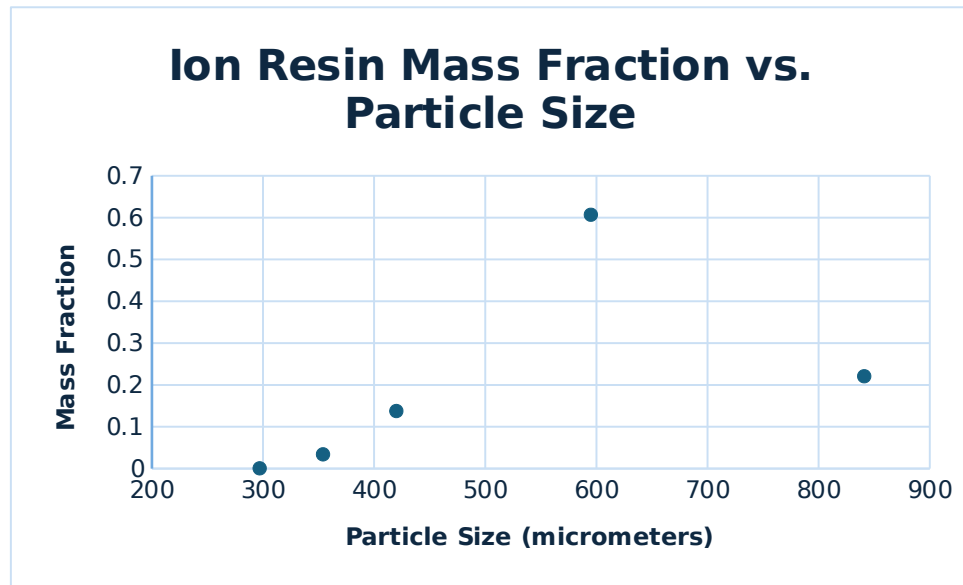


Figure 4- Ion Exchange Resin Mass Fraction vs. Particle Size

### Pressure Drop Behavior and Minimum Fluidization Velocity of Ion-Exchange Resin

Figures 5 and 6 show the pressure drop per unit bed height as a function of superficial gas velocity for the ion-exchange resin at bed heights of 6 inches and 10 inches. At low velocities, the pressure drop increases steadily with increasing gas velocity, which is characteristic of packed-bed behavior where the particles remain stationary and the gas flows through the void spaces in the bed.

As the gas velocity increases, the pressure drop reaches a maximum and then becomes nearly constant, indicating the onset of fluidization. This point corresponds to the minimum fluidization velocity, where the drag force exerted by the gas balances the effective weight of the particles. For the 6-inch resin bed, the minimum fluidization



velocity was identified at approximately **0.21 m/s**. A similar transition was observed for the 10-inch resin bed, where the minimum fluidization velocity was also approximately **0.21 m/s**. The similarity in values for both bed heights is expected, since minimum fluidization velocity depends primarily on particle and fluid properties rather than bed height.

A small hysteresis effect is observed between the increasing-flow and decreasing-flow curves, with slightly lower pressure drops measured during flow reduction. This behavior is commonly observed in fluidized beds and is attributed to particle rearrangement and changes in bed structure after fluidization has occurred.

The experimental pressure drop data were compared with predictions from the Ergun equation:

In this equation air density and viscosity was taken at ambient conditions . The ion-exchange resin was assumed to be nearly spherical, while the sand was assigned a lower sphericity consistent with rounded sand particles . The bed void fraction was initially estimated experimentally by filling a graduated cylinder with particles and adding water until the bed appeared fully wetted. However, this method produced inconsistent results and led to unrealistic pressure drop predictions when used in the Ergun model. As a result, representative void fraction values from literature were used for both materials, as these provided more reasonable agreement between the experimental data and model predictions [4].

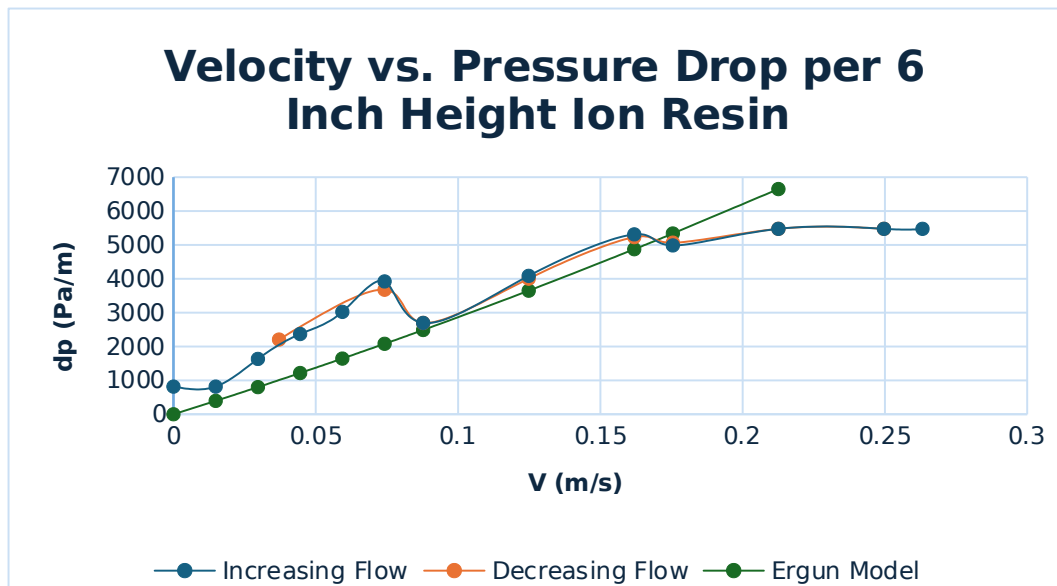


Figure 5- Velocity vs. Pressure Drop per 6 Inch Height of Ion-Exchange Resin

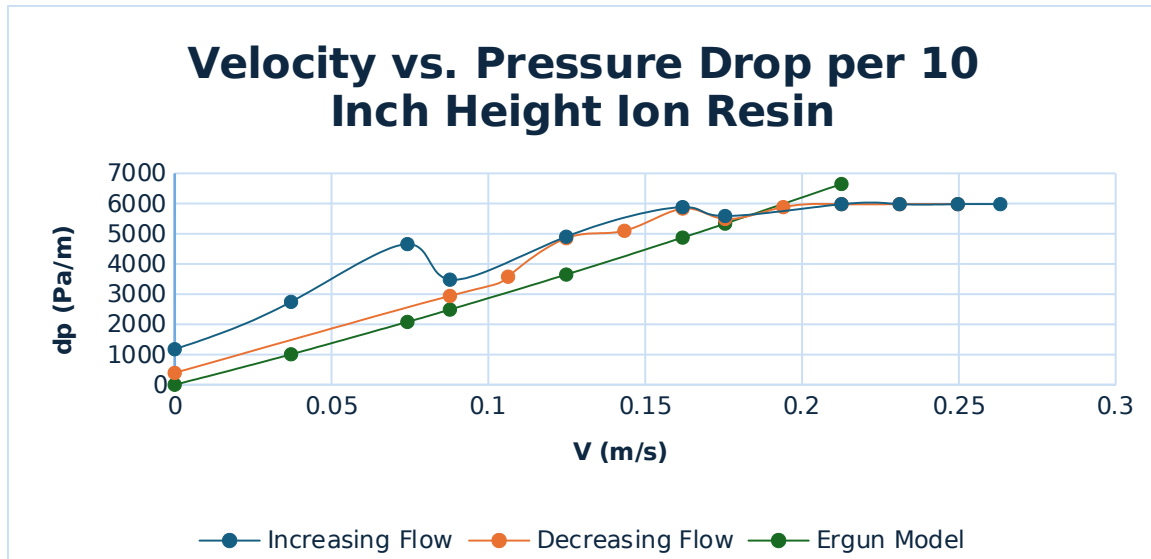


Figure 6- Velocity vs. Pressure Drop per 10 Inch Height of Ion-Exchange Resin

### Pressure Drop Behavior and Minimum Fluidization Velocity of Sand

Figures 7 and 8 show how the pressure drop per unit bed height changes with gas velocity for the sand bed at heights of 6 inches and 10 inches. At lower velocities, the pressure drop increases steadily as the flow rate increases. This behavior is expected and indicates that the bed is still packed, with the sand particles remaining stationary while the air flows through the void spaces between them.

As the gas velocity increases further, the pressure drop begins to level off and eventually reaches a nearly constant value. This flattening of the curve marks the onset of fluidization, where the upward force from the gas balances the weight of the sand particles. For the 6-inch sand bed, this transition occurs at a velocity of approximately **0.07 m/s**, which was taken as the experimental minimum fluidization velocity.

For the 10-inch sand bed, the transition to the plateau region is less sharp, making the minimum fluidization point more difficult to identify. Based on where the pressure drop becomes nearly constant, the minimum fluidization velocity was estimated to be approximately 0.09 m/s. The difference between the two bed heights reflects the gradual nature of sand fluidization and the fact that sand particles are irregular in shape and have a wider size distribution compared to the resin particles.

In both cases, a noticeable hysteresis is observed between the increasing-flow and decreasing-flow data, with lower pressure drops measured as the flow rate decreases. This behavior is typical for sand beds and is likely caused by particle rearrangement and friction effects once the bed has already been fluidized.

When compared with the Ergun equation, the model captures the general increase in pressure drop in the packed-bed region but tends to overpredict the pressure drop near the onset of fluidization. This difference is expected, since the Ergun equation assumes

uniform particle size, constant void fraction, and idealized particle shape, assumptions that are not fully satisfied for sand.

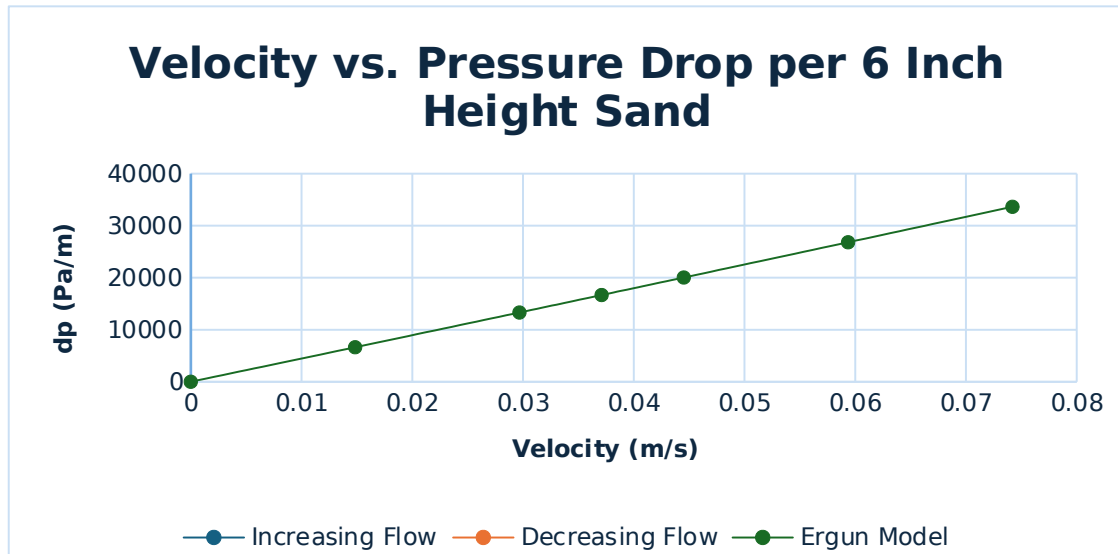


Figure 7- Velocity vs. Pressure Drop per 6 Inch Height of Sand

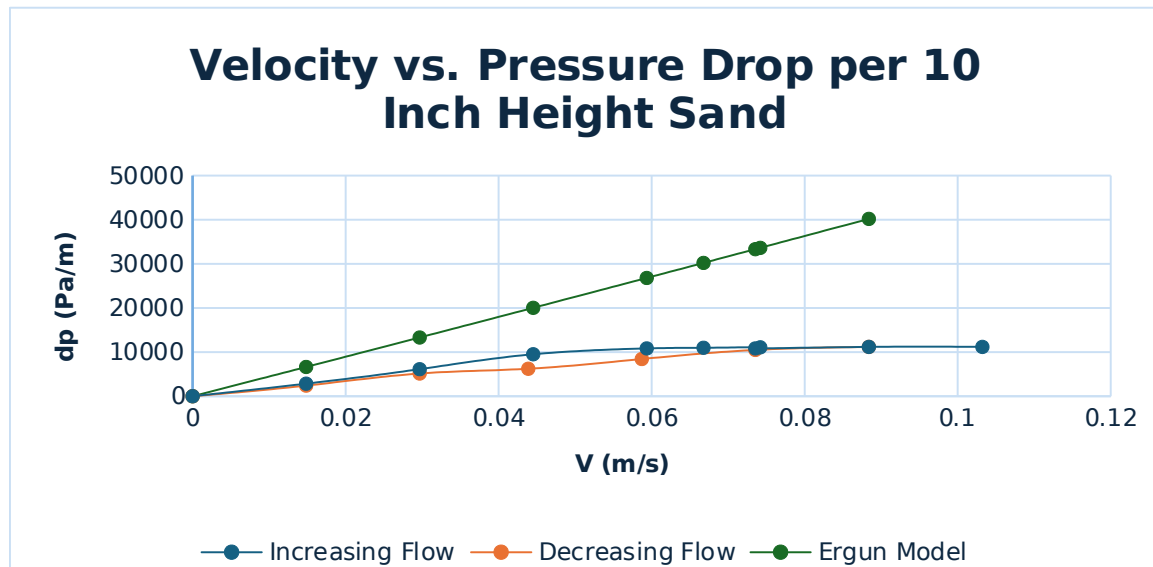


Figure 8- Velocity vs. Pressure Drop per 10 Inch Height of Sand

### Minimum Fluidization Velocity Calculation

The minimum fluidization velocity was also calculated using Equation (7.51) in McCabe et al. (2005), which is obtained by applying the Ergun equation at the point of incipient fluidization:

The values used in this calculation are summarized in Table X. Particle densities of 1200 kg/m<sup>3</sup> for the ion-exchange resin and 2650 kg/m<sup>3</sup> for sand were used [4]. Particle diameters were taken from the sieve analysis, while air density and viscosity were

assumed at ambient conditions. Particle sphericity values were selected from literature, and void fractions were taken from literature after experimental estimates produced inconsistent and unrealistic results.

Because Equation (7.51) is nonlinear in velocity, the minimum fluidization velocity was solved numerically using Goal Seek in Excel.

The calculated minimum fluidization velocities obtained from this method are shown in Table 1 along with the experimentally determined values. For the ion-exchange resin, the calculated minimum fluidization velocity of approximately 0.208 m/s agrees well with the experimental value of approximately 0.213 m/s, resulting in a percent difference of about 2% for both bed heights. In contrast, the calculated minimum fluidization velocity for sand was significantly lower than the experimental values, with larger percent differences observed. This discrepancy is attributed to the irregular shape and broad size distribution of sand particles, as well as uncertainty in the assumed void fraction and sphericity

*Table 2- Minimum Fluidization Velocity: Calculated vs. Experimental*

| Fluidization Velocity                            | Ionexchange Resin (height = 6 inches) | Ionexchange Resin (height = 10 inches) | Sand (height = 6 inches) | Sand (height = 10 inches) |
|--------------------------------------------------|---------------------------------------|----------------------------------------|--------------------------|---------------------------|
| Calculated Minimum Fluidization Velocity (m/s)   | 0.208                                 | 0.208                                  | 0.034                    | 0.034                     |
| Experimental Minimum Fluidization Velocity (m/s) | 0.213                                 | 0.213                                  | 0.074                    | 0.088                     |
| % Difference                                     | 2.4%                                  | 2.4%                                   | 121%                     | 163%                      |

Overall, this experiment studied the fluidization behavior of sand and ion-exchange resin by analyzing pressure drop as a function of gas velocity. Sieve analysis was used to determine particle sizes, which were then used to calculate and compare minimum fluidization velocities.

The ion-exchange resin showed good agreement between experimental and calculated minimum fluidization velocities, reflecting its relatively uniform size and near-spherical shape. In contrast, the sand exhibited a more gradual transition to fluidization and larger differences between experimental and calculated values, likely due to its irregular particle shape and broader size distribution. These results highlight the importance of particle properties and model assumptions when analyzing fluidized bed behavior.

## 5. Discussion (15 pts)

Overall, the pressure-drop trends matched the standard fluidized-bed behavior shown in the lab manual. Pressure drop rose in the fixed-bed region, reached a peak or plateau near incipient fluidization, then stayed roughly constant while bed expansion became the main change. The same general pattern was seen for both materials, and the difference between the increasing-flow and decreasing-flow curves followed the expected hysteresis behavior. This hysteresis means the bed does not return to the exact same packing after it has already been fluidized, so the pressure drop can be slightly lower on the way down. Below incipient fluidization, the fixed-bed region should follow packed-bed behavior, so it can be compared directly to the Ergun model. In general, the ion-exchange resin curves were “cleaner” with a sharper transition, while the sand curves were more gradual, which is consistent with the differences in particle shape and size distribution discussed below.

### Sieve Analysis and Particle Size Distribution

The sieve analysis confirmed that the two bed materials were in different particle-size ranges. The sand had an average particle diameter of 204.7  $\mu\text{m}$ , with most of the mass concentrated around the 200–250  $\mu\text{m}$  bins, indicating a fine material with a relatively narrow distribution. The ion-exchange resin had an average particle diameter of 585.3  $\mu\text{m}$ , and most of the mass was concentrated at larger sizes near 600  $\mu\text{m}$  with a broader spread toward the high end.

The resin particle size scale is consistent with common commercial ion-exchange resins, which are typically manufactured as nearly spherical beads with mean sizes on the order of  $\sim 0.5\text{--}0.7$  mm. These beads can span from a few hundred microns up to around a millimeter depending on grade, which agrees well with the resin distribution observed here and the mass concentration near  $\sim 600$   $\mu\text{m}$ . [7]

For the pressure-drop modeling, sphericity and void fraction were important assumptions. The resin was treated as nearly spherical with  $\phi \approx 1$ , which matches how these beads are made and also matches the sharper fluidization transition seen in the resin plots. The sand was assigned a lower sphericity of  $\phi \approx 0.83$  to reflect irregular grains, and this is one reason the sand data did not match the ideal-model trend as closely. Void fractions were initially estimated experimentally by wetting/fill methods, but those values were inconsistent and produced unrealistic pressure-drop predictions, so representative literature void fractions were used instead,  $\epsilon \approx 0.45$  for resin and  $\epsilon \approx 0.42$  for sand. This choice improved agreement in the fixed-bed region, which suggests the “representative  $\epsilon$ ” approach was more reliable than the quick experimental estimate for this setup.

### Pressure Drop and Minimum Fluidization Velocity of Ion-Exchange Resin

For the ion-exchange resin, both bed heights showed a clear transition from fixed-bed behavior to a near-plateau pressure drop, and the minimum fluidization velocity was  $\sim 0.21$  m/s for both 6 in and 10 in beds. That is the expected trend: in theory,  $U_{mf}$  depends mainly on particle and fluid properties, while bed height mainly scales the total pressure

drop. Since pressure drop was plotted per bed height here, the two resin curves collapse well and make the comparison clearer. The small hysteresis between increasing and decreasing flow also matches the typical loosening/repacking explanation for fluidized beds.

In the fixed-bed region below  $U_{mf}$ , the Ergun model captured the general increase in  $\Delta P/L$  with velocity reasonably well for the resin. That makes sense because the resin is closer to the Ergun assumptions: more uniform size, more spherical shape, and more stable packing. Near the transition region, the model tended to miss some features of the curve, especially around the approach to the plateau. This is expected because the Ergun equation assumes a packed bed with constant  $\epsilon$  and no bed expansion, while the actual system begins to form voids/bubbles and the packing structure changes as it approaches fluidization. Small instrument and setup effects can also matter more near  $U_{mf}$ , such as slight pressure fluctuations, non-uniform distributor flow, and the fact that the bed is not perfectly uniform across the tube cross-section.

As a rough comparison to other gas–solid beds, literature examples with polymeric/resin-type beads often report  $U_{mf}$  on the order of a few tenths of a m/s under ambient gas conditions, depending strongly on particle size and density. One reported case for resin-based beads in a gas–solid system gives  $U_{mf} \approx 0.26$  m/s with packed void fraction around 0.41 for a different resin-type material and size range. While not identical materials, it supports that the scale of the resin  $U_{mf}$  measured here ( $\sim 0.21$  m/s) is reasonable for gas fluidization of low-density, bead-like particles. [8]

### Pressure Drop and Minimum Fluidization Velocity of Sand

For sand, the behavior was still consistent with fluidization theory:  $\Delta P/L$  increased at low velocities and leveled off near incipient fluidization, but the transition was less sharp than the resin case. The measured minimum fluidization velocities were  $\sim 0.07$  m/s for the 6 in bed and  $\sim 0.09$  m/s for the 10 in bed. The larger uncertainty in identifying the transition point for sand is believable because the curve flattens more gradually, and sand has a broader size distribution and irregular shapes that smear out the “single”  $U_{mf}$  point into more of a transition region.

Compared to literature, sand  $U_{mf}$  values can vary widely with particle size distribution, sphericity, and the exact sand used. One published study explicitly states a sand  $U_{mf} \approx 0.026$  m/s and reports the same overall regime behavior as velocity increases, from fixed-bed conditions to incipient fluidization and then stronger motion at higher velocities. Although this value is lower than the  $U_{mf}$  measured here, it supports that our measured sand values are in a similar order of magnitude for gas fluidization, and differences are reasonable since  $U_{mf}$  is very sensitive to the effective particle diameter, shape, and packing behavior. [9]

The Ergun model overpredicted the sand pressure drop in the fixed-bed region and especially near the onset of fluidization, as seen by the model curve sitting well above the experimental curves. This likely comes from the fact that sand is harder to represent

using a single “effective” diameter and a single packing structure, so the real bed can deviate from the ideal packed-bed assumptions used in Ergun. In addition, the void fraction is not truly constant as the bed loosens approaching  $U_{mf}$ , and minor flow maldistribution through the distributor can reduce the measured  $\Delta P$  compared to the uniform-flow assumption. These effects become more noticeable near incipient fluidization, which is why the mismatch is strongest in that region.

**6. References (10 pts)**

- [1] Department of Chemical Engineering. *Fluid Flow Through Fluidized Bed*. University of Illinois at Chicago, 2025.
- [2] Bird, R., Stewart, W. and Lightfoot, E. (2002) *Transport Phenomena*. 2nd Edition, John Wiley and Sons, New York.
- [3] McCabe W.L., Smith J.C., Harriot P., “Unit Operations in Chemical Engineering” 7th Edition pp. 882-922, (2005)
- [4] Green, D. W., & Perry, R. H. (Eds.). (2008). *Perry’s chemical engineers’ handbook* (8th ed.). McGraw-Hill.
- [5] National Aeronautics and Space Administration. (n.d.). Air mass/density. NASA Earthdata.<https://earthdata.nasa.gov/topics/atmosphere/air-mass-density>
- [6] Engineers Edge. (n.d.). Viscosity of air: Dynamic and kinematic.[https://www.engineersedge.com/physics/viscosity\\_of\\_air\\_dynamic\\_and\\_kinematic\\_14483.htm](https://www.engineersedge.com/physics/viscosity_of_air_dynamic_and_kinematic_14483.htm)
- [7] DuPont Water Solutions. *AmberLite™ Ion Exchange Resins: Product Data Sheet* (includes typical bead size / particle size distribution). Accessed February 1, 2026.
- [8] Sjöström, E., et al. “The Effect of Humidity on the Fluidization Properties of Powder and Granular CO<sub>2</sub> Sorbents.” *International Journal of Greenhouse Gas Control* (year varies by version/record; used for reported U<sub>mf</sub> scale and packed void fraction example for bead-like granular sorbents).
- [9] Cáceres-Martínez, Louis Edwards, Diana Carolina Guío-Pérez, and Sonia Lucía Rincón-Prat. “Significance of the Particle Physical Properties and the Geldart Group in the Use of Correlations for the Prediction of Minimum Fluidization Velocity of Biomass–Sand Binary Mixtures.” *Biomass Conversion and Biorefinery* (published online 2021; in print 2023).



**7. Appendix I: Data Tabulation/Graphs (10 pts)**

The tables below display air flow rate and velocity against pressure drop determined experimentally and pressure drop calculated using the Ergun equation for both material beds ion-exchange resin and sand per 6-inch and 10-inch height of material.

*Table 3- Flow Rate vs Pressure Drop per 6 Inch Ion-Exchange Bed*

| Flow(LPM) | dP(inchwater) |
|-----------|---------------|
| 0         | 0.5           |
| 7.218     | 0.5           |
| 14.436    | 1             |
| 21.654    | 1.45          |
| 28.872    | 1.85          |
| 36.09     | 2.4           |
| 42.68     | 1.65          |
| 60.725    | 2.5           |
| 78.77     | 3.25          |
| 85.36     | 3.05          |
| 103.405   | 3.35          |
| 121.45    | 3.35          |
| 128.04    | 3.35          |
| 121.45    | 3.35          |
| 103.405   | 3.35          |
| 85.36     | 3.1           |
| 78.77     | 3.2           |
| 60.725    | 2.45          |
| 42.68     | 1.65          |
| 36.09     | 2.25          |
| 18.045    | 1.35          |

Table 4- Velocity vs Pressure Drop per 6 Inch Ion-Exchange Bed

| V(m/s) | dP (Pa/m) | ErgundP (Pa/m) |
|--------|-----------|----------------|
| 0.00   | 816.93    | 0.00           |
| 0.01   | 816.93    | 395.56         |
| 0.03   | 1633.86   | 801.39         |
| 0.04   | 2369.09   | 1217.50        |
| 0.06   | 3022.64   | 1643.88        |
| 0.07   | 3921.26   | 2080.54        |
| 0.09   | 2695.87   | 2488.18        |
| 0.12   | 4084.65   | 3648.24        |
| 0.16   | 5310.04   | 4872.52        |
| 0.18   | 4983.27   | 5335.64        |
| 0.21   | 5473.43   | 6647.60        |
| 0.25   | 5473.43   |                |
| 0.26   | 5473.43   |                |
| 0.25   | 5473.43   |                |
| 0.21   | 5473.43   |                |
| 0.18   | 5064.96   |                |
| 0.16   | 5228.35   |                |
| 0.12   | 4002.95   |                |
| 0.09   | 2695.87   |                |
| 0.07   | 3676.18   |                |
| 0.04   | 2205.71   |                |

Table 5- Flow Rate vs. Pressure Drop per 10 Inch Ion-Exchange Bed

| Flow(LPM) | dP(inchwater) |
|-----------|---------------|
| 0         | 1.2           |
| 18.045    | 2.8           |
| 36.09     | 4.75          |
| 42.68     | 3.55          |
| 60.725    | 5             |
| 78.77     | 6             |
| 85.36     | 5.7           |
| 103.405   | 6.1           |
| 112.4275  | 6.1           |
| 121.45    | 6.1           |
| 128.04    | 6.1           |
| 121.45    | 6.1           |
| 112.4275  | 6.1           |
| 103.405   | 6.1           |
| 94.3825   | 6             |
| 85.36     | 5.6           |
| 78.77     | 5.95          |
| 69.7475   | 5.2           |
| 60.725    | 4.95          |
| 51.7025   | 3.65          |
| 42.68     | 3             |
| 0         | 0.4           |

Table 6- Velocity vs Pressure Drop per 10 Inch Ion-Exchange Bed

| V(m/s) | dp (Pa/m) | Ergundp (Pa/m) |
|--------|-----------|----------------|
| 0.00   | 1176.38   | 0.00           |
| 0.04   | 2744.88   | 1008.16        |
| 0.07   | 4656.50   | 2080.54        |
| 0.09   | 3480.12   | 2488.18        |
| 0.12   | 4901.57   | 3648.24        |
| 0.16   | 5881.89   | 4872.52        |
| 0.18   | 5587.80   | 5335.64        |
| 0.21   | 5979.92   | 6647.60        |
| 0.23   | 5979.92   |                |
| 0.25   | 5979.92   |                |
| 0.26   | 5979.92   |                |
| 0.25   | 5979.92   |                |
| 0.23   | 5979.92   |                |
| 0.21   | 5979.92   |                |
| 0.19   | 5881.89   |                |
| 0.18   | 5489.76   |                |
| 0.16   | 5832.87   |                |
| 0.14   | 5097.64   |                |
| 0.12   | 4852.56   |                |
| 0.11   | 3578.15   |                |
| 0.09   | 2940.94   |                |
| 0.00   | 392.13    |                |

*Table 7- Flow Rate vs. Pressure Drop per 6 Inch Height Sand*

| Flow(LPM) | dP(inchwater) |
|-----------|---------------|
| 0         | 0             |
| 7.218     | 1.6           |
| 14.436    | 3.4           |
| 18.045    | 4.4           |
| 21.654    | 5.3           |
| 28.872    | 5.75          |
| 36.09     | 6.05          |
| 35.776    | 5.9           |
| 42.994    | 6             |
| 50.212    | 6.05          |
| 57.43     | 6.05          |
| 60.725    | 6.05          |
| 57.43     | 6.05          |
| 50.212    | 6.05          |
| 42.994    | 6             |
| 35.776    | 5.75          |
| 28.558    | 4.65          |
| 21.34     | 3.4           |
| 21.654    | 4.65          |
| 14.436    | 2.9           |
| 7.218     | 1.45          |
| 0         | 0             |

*Table 8- Flow Rate vs. Pressure Drop per 10 Inch Height Sand*

| Flow(LPM) | dP(inchwater) |
|-----------|---------------|
| 0         | 0             |
| 7.218     | 2.9           |
| 14.436    | 6.25          |
| 21.654    | 9.7           |
| 28.872    | 11.05         |
| 32.481    | 11.2          |
| 36.09     | 11.3          |
| 35.776    | 11.05         |
| 42.994    | 11.4          |
| 50.212    | 11.4          |
| 42.994    | 11.4          |
| 35.776    | 10.7          |
| 28.558    | 8.6           |
| 21.34     | 6.35          |
| 21.654    | 8.1           |
| 14.436    | 5.25          |
| 7.218     | 2.45          |
| 0         | 0             |

Table 9- Velocity vs. Pressure Drop per 6 Inch Height Sand

| V(m/s) | dP (Pa/m) | ErgundP (Pa/m) |
|--------|-----------|----------------|
| 0.00   | 0.00      | 0.00           |
| 0.01   | 2614.17   | 6630.53        |
| 0.03   | 5555.12   | 13307.94       |
| 0.04   | 7188.98   | 16664.23       |
| 0.04   | 8659.45   | 20032.23       |
| 0.06   | 9394.69   | 26803.41       |
| 0.07   | 9884.84   | 33621.46       |
| 0.12   | 9884.84   |                |
| 0.12   | 9884.84   |                |
| 0.12   | 9884.84   |                |
| 0.10   | 9884.84   |                |
| 0.09   | 9803.15   |                |
| 0.07   | 9394.69   |                |
| 0.06   | 7597.44   |                |
| 0.04   | 5555.12   |                |
| 0.03   | 4738.19   |                |
| 0.01   | 2369.09   |                |
| 0.00   | 0.00      |                |

Table 10- Velocity vs. Pressure Drop per 10 Inch Height Sand

| V(m/s) | dp (Pa/m) | Ergun dp (Pa/m) |
|--------|-----------|-----------------|
| 0.00   | 0.00      | 0.00            |
| 0.01   | 2842.91   | 6630.53         |
| 0.03   | 6126.97   | 13307.94        |
| 0.04   | 9509.06   | 20032.23        |
| 0.06   | 10832.48  | 26803.41        |
| 0.07   | 10979.53  | 30206.58        |
| 0.07   | 11077.56  | 33621.46        |
| 0.07   | 10832.48  | 33323.89        |
| 0.09   | 11175.59  | 40186.78        |
| 0.10   | 11175.59  |                 |
| 0.09   | 11175.59  |                 |
| 0.07   | 10489.37  |                 |
| 0.06   | 8430.71   |                 |
| 0.04   | 6225.00   |                 |
| 0.03   | 5146.65   |                 |
| 0.01   | 2401.77   |                 |
| 0.00   | 0.00      |                 |

**8. Appendix II: Error Analysis (10 pts)**

There were some sources of error that might have affected the results of this experiment. One source of error might have come from reading the manometer which was used to measure the pressure drop across the bed, and since it was not digital it was harder to get accurate readings since there were small fluctuations in the fluid level and the smallest readable increment on the scale was approximately 0.05 inches of water, which corresponds to about  $\pm 12$  Pa. The difficulty reading the scale may have caused slight inaccuracies in the pressure drop values since the minimum fluidization velocity was determined from pressure drop, these measurement errors could have affected the exact point at which fluidization occurred. In addition, the flow rate was adjusted manually using increments in LPM, which limited the accuracy of the air velocity measurements. Another source of error came from switching between the two flowmeters during the experiment, which caused brief jumps in the measured pressure drop as the airflow was redistributed through the system.

An additional source of error probably came from particles themselves since both the sand and ion-exchange resin had a range of particle sizes average particle diameter of  $204.7\text{ }\mu\text{m}$  for sand and  $585.3\text{ }\mu\text{m}$  for ion-exchange resin which could affect how air flowed through the bed and influence the measured pressure drop. The bed was also packed differently every time, even with an attempt to make them packed the same way they still ended up being different, which could have affected the results. In addition, we might have taken some measurements before the system reached a steady state, especially when the bed first started to fluidize, where the pressure readings were harder to read and varied by about 0.05 inches of water. All those errors together could contribute to uncertainty in the pressure drop measurements and the determination of the minimum fluidization velocity. Also, the walls of the column most likely affected how particles moved near the edges of the bed, especially at lower air velocities, which could have slightly affected the pressure drop measurements.

Overall, the results of this experiment could have been affected by the sources of error mentioned above, which could cause uncertainty. Measurement limitations in reading the airflow rates with variations in particle size and bed packing could also cause uncertainty in the pressure drop measurements. Factors like wall effects and data collection before reaching steady state conditions could have influenced the results. While those errors probably affected the exact values of the minimum fluidization velocity, overall trends observed in the experiment were still consistent with expected fluidization behavior.

**9. Appendix III: Sample Calculations (10 pts)**

Most of the calculations in this experiment revolved around the Ergun Equation. The following calculation is for a set of values found from 6 inches of ion exchange resin packed inside the column:

**10. Appendix IV: Individual Team Contributions**

|                                           | <b>Time (Hours)</b>   |                   |                        |                        |
|-------------------------------------------|-----------------------|-------------------|------------------------|------------------------|
|                                           | <b>Joshua Olusoji</b> | <b>Omar Zaruk</b> | <b>Sanad Abu Hweij</b> | <b>Qussay Suleiman</b> |
| <b>Operator (Both Lab Days)</b>           | 4                     | 5                 | 4                      | 3                      |
| <b>Pre-Lab Editing</b>                    | 1                     | 1                 | 2                      | 2                      |
| <b>Experimental Objective (Prelab)</b>    |                       |                   |                        | 1                      |
| <b>Apparatus (prelab)</b>                 | 2                     |                   |                        |                        |
| <b>Materials &amp; Supplies (prelab)</b>  |                       |                   | 1                      |                        |
| <b>Procedure (prelab)</b>                 |                       | 1                 |                        |                        |
| <b>Project Safety Evaluation (prelab)</b> |                       |                   |                        | 2                      |
| <b>Final Lab Editing</b>                  | 1                     | 1                 |                        | 1                      |
| <b>Executive Summary (final lab)</b>      |                       | 1                 |                        | 1                      |
| <b>Introduction (Final lab)</b>           |                       |                   | 3                      |                        |
| <b>Theory (Final Lab)</b>                 | 2                     |                   |                        |                        |
| <b>Results (final lab)</b>                |                       | 7                 |                        |                        |
| <b>Discussion (final lab)</b>             |                       |                   |                        | 4                      |
| <b>References (final and/or prelab)</b>   | 1                     | 1                 |                        | 0.5                    |
| <b>Data Tabulation/Graphs (final)</b>     |                       | 1                 |                        |                        |
| <b>Error Analysis (final lab)</b>         | 2                     |                   | 3                      |                        |
| <b>Sample Calculations (final Lab)</b>    |                       |                   |                        |                        |
| <b>Power Point Presentation</b>           | 2                     |                   | 1.5                    | 2                      |
| <b>Total</b>                              | 15                    | 18                | 14.5                   | 16.5                   |